The Ghost in the Genome: High-Dimensional Constraints and the Bi-state Dynamics of Homologous Gene Groups

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Abstract

The organization and evolution of homologous genes are commonly analyzed at the level of individual genes and sequence similarity. However, extensive sequence divergence and alignment uncertainty increasingly limit our ability to characterize deep evolutionary relationships and family-level structure. Here, we propose a conceptual framework in which homologous gene groups are treated as fundamental evolutionary entities, rather than as collections of independently evolving genes. We formulate three testable hypotheses. First, homologous gene groups follow a bi-state evolutionary dynamic, comprising a long-lasting stable state in which group presence is conserved despite continuous internal reorganization, and a transition state associated with the emergence or loss of entire groups. Second, the long-term organization of homologous gene groups is governed by concurrent evolutionary constraints acting at multiple levels, including gene-level, group-level and genome-ecological contexts, such that family-level structure cannot be explained by sequence-level constraints alone. Third, high-dimensional protein representations learned by protein language models reflect latent similarities induced by these shared multi-layer evolutionary constraints and therefore organize proteins according to group-level structure even in the remote-homology regime. This framework provides a unified interpretation of homologous gene group organization, evolutionary dynamics and representation-based analysis, and clarifies why recent representation learning approaches succeed where classical alignment-based methods become unreliable. Our perspective establishes a testable theoretical basis for integrating comparative genomics, gene family evolution and protein representation learning.

Keywords:

homologous gene groups; bi-state evolutionary dynamics; multi-layer evolutionary constraints; protein language models; representation space; remote homology

1. Introduction

Understanding how proteins evolve and diversify remains a central challenge in evolutionary genomics, particularly for large and complex gene families in which functional conservation persists despite extensive sequence divergence. While alignment-based methods and phylogenetic reconstruction have provided a powerful framework for comparative analysis, they offer only a limited view of the organizational principles that govern gene family evolution. Here, we propose a conceptual framework that redefines homologous gene groups as explicit evolutionary entities and describes their long-term dynamics through bi-state evolutionary behavior and multi-layer evolutionary constraints. This perspective provides a unifying theoretical foundation for interpreting recent advances in representation-based protein models and for extending comparative genomics beyond the practical limits of sequence similarity.

From alignment to representation: why classical homology theory is no longer sufficient

Comparative genomics has long been built upon the alignment paradigm, in which evolutionary relatedness and functional conservation are inferred primarily from sequence similarity. This paradigm has been remarkably successful for closely related genes and species, and remains the foundation of most gene family identification and phylogenetic analyses. However, an increasing number of studies have revealed a persistent and systematic limitation: a substantial fraction of homologous proteins retain similar structural and functional properties despite exhibiting extremely low sequence similarity, commonly referred to as remote homology [1,2].

This long-standing "twilight zone" problem is not merely a technical limitation of alignment algorithms, but reflects a deeper theoretical issue. Sequence alignment implicitly assumes that the evolutionary information relevant to structure and function is sufficiently encoded in local or global residue similarity. Yet evolutionary constraints acting on proteins arise from multiple sources, including physicochemical requirements of folding and activity, dosage balance and functional complementarity within gene families, and large-scale ecological and genomic contexts. These constraints do not necessarily translate into simple residue-level conservation patterns, especially over long evolutionary timescales.

Recent advances in protein language models and structure prediction systems have further exposed this conceptual gap [3,4]. Without relying on explicit sequence alignment, these models are able to capture functional and structural relationships across highly divergent proteins. Their empirical success highlights a critical question that current evolutionary theory does not adequately address: what underlying biological regularities are being learned when sequence similarity alone is no longer informative [5,6]?

In this Perspective, we argue that this gap arises from an implicit mismatch between the true objects of evolutionary organization and the units typically analyzed in comparative genomics. We propose that a shift in perspective is required, from individual genes viewed in isolation to homologous gene groups as higher-level evolutionary entities, and from alignment-based comparison to representation-based characterization of evolutionary constraints. On this basis, we introduce a unified theoretical framework centered on homologous gene groups, bi-state evolutionary dynamics and multi-layer evolutionary constraints, providing a conceptual foundation for understanding why high-dimensional protein representations are able to recover evolutionary relationships that escape classical alignment-based approaches.

Three testable hypotheses

To make the proposed conceptual framework explicit and falsifiable, we formulate the following three testable hypotheses concerning the organization and evolution of homologous gene groups.

Hypothesis 1: Bi-state evolutionary dynamics of homologous gene groups

Homologous gene groups exhibit a bi-state evolutionary dynamic, consisting of (i) a long-lasting stable state in which the presence of a gene group is conserved across lineages while its internal member composition is continuously reorganized, and (ii) a transition state in which entire gene groups emerge or disappear in association with major genomic or ecological shifts.

Testability

This hypothesis can be tested by reconstructing the evolutionary histories of homologous gene groups across multiple lineages and timescales, and by quantifying the relative stability of group presence versus the turnover and reorganization of member genes. Specifically, group-level presence-absence patterns can be compared with internal duplication-loss histories and subgroup rearrangements to determine whether group persistence is decoupled from member-level dynamics.

Hypothesis 2: Dominance of multi-layer evolutionary constraints at the group level

The long-term organization and diversification of homologous gene groups are governed by concurrent evolutionary constraints acting at multiple levels, including the gene level, the group level and the genome-ecological context, such that family-level structure cannot be fully explained by sequence-level constraints alone.

Testability

This hypothesis can be evaluated by assessing whether models that integrate sequence-derived features with group-level properties and lineage-specific contextual variables better explain observed family-level patterns — such as subgroup organization, functional complementarity and copy-number structure—than models relying solely on sequence similarity or site-wise evolutionary constraints.

Hypothesis 3: Representation space reflects shared evolutionary constraints

High-dimensional protein representations learned by protein language models capture latent similarities induced by shared multi-layer evolutionary constraints, and therefore organize proteins according to family-level and group-level structure even when classical sequence similarity is no longer informative.

Testability

This hypothesis can be tested by comparing neighborhoods or clusters in representation space with independently inferred group membership, subgroup structure and phylogenetic relationships. In particular, one can evaluate whether distances in representation space predict homologous group affiliation, internal organization and cross-species correspondence more consistently than alignment-based similarity in the remote-homology regime.

2. What is the real evolutionary entity?

Homologous gene groups as lineage-independent evolutionary units

In most comparative genomic analyses, the gene is implicitly treated as the basic evolutionary unit, and evolutionary relationships are described by mapping individual genes onto a species tree [7]. Gene families are usually introduced as convenient organizational categories rather than as explicit evolutionary entities. This practice obscures an important distinction between the behavior of individual gene copies and the evolutionary dynamics of the family as a whole.

Here we define a homologous gene group (HGG) as a set of genes that originate from a single ancestral gene and share a common evolutionary origin, regardless of their current species distribution. In practice, an HGG corresponds closely to a rigorously delimited gene family, but the concept emphasizes its ontological status as an evolutionary unit rather than a purely classificatory label.

Under this definition, an HGG possesses a lineage history that is not confined to any single species lineage. Its evolutionary trajectory is shaped jointly by gene duplication, loss and functional divergence events occurring across multiple species and multiple evolutionary timescales [8,9]. Individual gene copies represent transient realizations of this higher-level entity, whereas the HGG itself persists as a structured evolutionary object.

Importantly, the evolutionary properties of an HGG cannot be fully reduced to the properties of its individual members. The number of members, their functional complementarity and their collective dosage balance are maintained and reshaped by selective pressures that act at the level of the group [10]. Consequently, treating homologous genes solely as independent units attached to a species tree conflates copy-level processes with group-level organization.

In the following sections, we therefore treat homologous gene groups as lineage-independent evolutionary entities. This shift in evolutionary ontology provides the conceptual basis for describing family-level dynamics, for formulating bi-state evolutionary behavior, and for analyzing how higher-order constraints shape the long-term organization of complex gene families.

3. Bi-state evolutionary dynamics of homologous gene groups

Once homologous gene groups are treated as explicit evolutionary entities, their long-term behavior can no longer be adequately described as a continuous and homogeneous process. Instead, we propose that the evolutionary dynamics of homologous gene groups follow a characteristic bi-state pattern, alternating between a stable phase and a transition phase.

In the stable phase, the overall number of homologous gene groups in a genome or lineage remains effectively constant. During this phase, evolutionary change is dominated by internal reorganization within each group. Gene duplication and loss events may occur, but they are largely compensated, resulting in relatively stable group-level presence. Functional differentiation, subfunctionalization and fine-scale regulatory divergence primarily take place among existing members. Consequently, the main evolutionary activity is expressed as changes in the internal structure of the group rather than the creation or elimination of groups themselves.

In contrast, during the transition phase, the organization of homologous gene groups becomes unstable. The number of gene groups may change due to the emergence of new groups from independent ancestral origins or the complete extinction of existing groups through the loss of all their members. Such phases are typically associated with major environmental shifts, large-scale genomic rearrangements or whole-genome duplication events, which alter the selective landscape and enable rapid restructuring of gene family repertoires.

A key distinction in this framework is the separation between group number dynamics and member number dynamics. In the stable phase, the set of homologous gene groups is approximately conserved, whereas the number and functional composition of members within each group remain dynamic. In the transition phase, both the number of groups and the internal composition of groups may change abruptly. This distinction is critical, because traditional gene family analyses often conflate variation in copy number with changes in the evolutionary status of the group itself.

Under short evolutionary time intervals, homologous gene groups can therefore be approximated as a finite and stable set of evolutionary entities. Over longer timescales, however, the system exhibits punctuated reorganization, in which the appearance and disappearance of entire groups reshapes the functional landscape of genomes. This bi-state dynamic provides a unifying description for how gene families can simultaneously display long-term continuity and episodic innovation.

Importantly, this formulation does not assume that all groups undergo transitions synchronously. Instead, different homologous gene groups may enter transition phases independently, depending on their functional roles and selective contexts. The bi-state model thus captures both the global stability of gene family repertoires and the localized, group-specific bursts of evolutionary change that characterize the diversification of complex gene families.

4. Multi-layer evolutionary constraints shaping gene families

The bi-state dynamics of homologous gene groups implies that evolutionary change is not governed by a single, uniform selection mechanism. Instead, the long-term organization and diversification of gene families are shaped by multiple layers of evolutionary constraints acting simultaneously on different organizational levels.

At the gene level, evolutionary change is constrained by fundamental physicochemical requirements. Amino-acid substitutions must preserve basic folding stability, catalytic capacity and molecular interactions. These constraints operate on individual protein sequences and define a minimal feasible sequence space for each gene product [11].

At the group level, additional constraints arise from the internal organization of homologous gene groups. Members of the same group are not functionally independent. Their collective composition must maintain functional complementarity, regulatory coordination and dosage balance. Duplication, loss and functional divergence of individual members therefore affect not only single genes but also the overall structure and functional coverage of the group. Selection at this level acts to preserve a coherent internal organization of the group, even when individual sequences diverge substantially.

At the global level, homologous gene groups are further constrained by species-specific ecological conditions and genome-wide architectural contexts. Environmental pressures, life-history strategies and large-scale genomic events influence which groups are retained, expanded or eliminated across evolutionary lineages. These constraints act on the presence and relative importance of entire groups rather than on individual gene copies.

These three layers of constraints operate concurrently rather than hierarchically. A sequence change that is permissible at the gene level may be disfavored at the group level if it disrupts functional complementarity, and a group configuration that is locally optimal may still be eliminated at the global level under changing ecological or genomic conditions. Consequently, evolutionary outcomes observed in gene families reflect the combined effects of physicochemical feasibility, group-level organization and lineage-specific selective environments.

This multi-layer constraint framework provides a natural explanation for why extensive sequence divergence can coexist with preserved family-level structure and conserved functional roles. It also establishes the conceptual basis for interpreting gene family evolution as a high-dimensional process, in which observable sequences represent only one projection of a more complex constrained evolutionary system.

5. Proteins as high-dimensional evolutionary entities

Traditional molecular descriptions treat protein sequences, three-dimensional structures and biological functions as distinct but related properties. In most evolutionary analyses, sequence variation is regarded as the primary observable, while structure and function are interpreted as downstream consequences. This implicit hierarchy, however, becomes increasingly insufficient when remote homologs exhibit strong functional and structural similarity despite extensive sequence divergence [12].

Here we propose a different conceptual view. A protein should not be regarded as a low-dimensional object defined solely by its primary sequence. Instead, it represents a realization of a high-dimensional evolutionary entity that is jointly constrained by multiple layers of evolutionary forces. The amino-acid sequence, the folded structure and the observed biological function correspond to different observable projections of this underlying entity.

From this perspective, the primary sequence reflects only one projection of the constrained evolutionary space, shaped primarily by physicochemical feasibility at the gene level. Protein structure represents another projection that integrates both local stability constraints and higher-order interaction patterns. Biological function, in turn, reflects how this constrained entity operates within cellular and organismal systems, and is influenced by group-level organization and ecological context.

Importantly, the mapping between these projections is neither linear nor one-to-one. Different sequences may correspond to similar regions of the underlying constrained space and therefore give rise to similar structures and functions. Conversely, small changes in sequence can sometimes induce large functional shifts when they alter how a protein occupies this constrained space within a gene family or a regulatory network.

This view naturally explains why sequence-based similarity metrics can fail to capture meaningful evolutionary relationships, particularly among remote homologs. Sequence divergence reflects only variation along one observable dimension, whereas evolutionary conservation is governed by constraints acting in a much higherdimensional space defined jointly by molecular stability, family-level organization and lineage-specific selective environments.

By treating proteins as projections of high-dimensional evolutionary entities, evolutionary analysis can move beyond the exclusive reliance on pairwise sequence comparison and instead focus on representations that better capture the structure of the underlying constrained evolutionary space.

6. Why protein language models work

PLMs as approximations of evolutionary manifolds

The conceptual framework outlined above naturally raises a central question: why are protein language models able to recover structural and functional relationships among highly divergent sequences that escape classical alignment-based approaches?

From the perspective of homologous gene groups and multi-layer evolutionary constraints, the success of protein language models can be interpreted as a consequence of how evolutionary information is implicitly distributed across sequences. The evolutionary regularities that govern protein variation are not confined to individual positions or local motifs, but are shaped by long-range dependencies, family-level organization and lineage-specific selective contexts. These regularities collectively define a constrained, high-dimensional evolutionary space.

Protein language models learn statistical representations from large collections of protein sequences without explicit supervision or predefined evolutionary models. In doing so, they implicitly capture patterns of residue co-variation, context-dependent substitution preferences and long-range dependencies that arise from shared evolutionary constraints. Importantly, these patterns are not limited to local sequence similarity and therefore do not require explicit sequence alignment to be detected.

Within the present framework, the embedding space learned by protein language models can be interpreted as a numerical approximation of the underlying constrained evolutionary space defined by gene-level, group-level and global constraints. Rather than encoding evolutionary relationships through explicit ancestry reconstruction, these models provide a continuous representation in which proximity reflects similarity with respect to shared evolutionary constraints.

This interpretation does not imply that protein language models reconstruct phylogenetic histories or explicitly identify homologous gene groups. Instead, they offer a representation in which proteins shaped by similar combinations of evolutionary constraints tend to occupy nearby regions of the embedding space, even when their primary sequences are highly divergent. In this sense, protein language models provide access to a latent structure of evolutionary organization that is only weakly reflected in sequence identity alone.

Crucially, this view does not position representation-based approaches as replacements for evolutionary inference. Rather, they should be regarded as complementary tools that capture aspects of evolutionary organization that are difficult to formalize within traditional alignment-based or tree-based frameworks. Protein language models therefore become valuable not because they bypass evolutionary theory, but because they offer an alternative numerical projection of the same high-dimensional evolutionary constraints that shape homologous gene groups over long evolutionary timescales.

7. Implications for comparative genomics and evolutionary biology

Reframing homologous gene groups as evolutionary entities and interpreting protein variation as projections of high-dimensional constrained spaces has direct implications for how comparative genomics and evolutionary analyses are conducted.

First, gene family analysis can no longer be viewed solely as an auxiliary step for phylogenetic reconstruction. Instead, family-level organization becomes a primary object of evolutionary inference. The internal structure of homologous gene groups, including the composition and arrangement of functional subgroups, provides information that is not fully captured by gene trees derived from multiple sequence alignments.

Second, this framework clarifies the conceptual role of representation-based analyses in evolutionary studies. When evolutionary constraints are distributed across multiple organizational levels and expressed through complex, high-dimensional dependencies, it becomes unrealistic to expect that all informative signals can be recovered through explicit alignment and site-wise comparison. Representation-based approaches provide a complementary view in which proteins and gene family members are compared in terms of shared constraint patterns rather than shared residues.

Third, classical phylogenetic methods retain an essential but more specialized role. Phylogenetic trees remain indispensable for reconstructing ancestry, timing and lineage relationships. However, they describe only one projection of evolutionary organization — namely, the historical branching structure inferred from aligned sequences. In the presence of extensive sequence divergence, functional reorganization and family-level constraints, phylogenetic trees alone are insufficient to describe the full structure of gene family evolution.

Within this perspective, representation-based analyses and alignment-based phylogenetics address different but complementary questions. Phylogenetic inference focuses on lineage history, whereas representation-based characterization emphasizes constraint similarity and organizational structure. Integrating these two views enables a more complete description of how homologous gene groups diversify, reorganize and maintain functional coherence over long evolutionary timescales.

Finally, treating gene families as structured evolutionary entities opens new possibilities for comparative studies across distantly related species. When proteins are compared in a unified representational space, family-level relationships can be examined even when sequence similarity falls below the practical limits of alignment-based approaches. This provides a conceptual foundation for extending comparative genomics beyond the traditional boundaries imposed by sequence divergence.

8. Conclusion and outlook

In this Perspective, we propose a unified theoretical framework for understanding the evolution of complex gene families by redefining the fundamental evolutionary unit as the homologous gene group. By introducing bi-state evolutionary dynamics and multi-layer evolutionary constraints, we provide a coherent description of how gene families preserve internal organization while undergoing long-term functional diversification and episodic structural reconfiguration.

Within this framework, proteins are no longer treated as low-dimensional sequence objects, but as observable projections of high-dimensional evolutionary entities shaped jointly by physicochemical feasibility, group-level organizational constraints and lineage-specific ecological and genomic contexts. This view offers a natural explanation for the long-standing discrepancy between extensive sequence divergence and conserved structural or functional properties among remote homologs.

We further argue that protein language models provide a computational approximation of this hidden constrained evolutionary space. Rather than replacing evolutionary theory, representation-based models offer an alternative numerical projection of the same multi-layer constraints that govern the organization and diversification of homologous gene groups. This perspective clarifies why such models are capable of capturing relationships that are only weakly reflected in pairwise sequence similarity.

Importantly, the framework presented here does not seek to replace classical phylogenetic analysis. Instead, it repositions phylogenetic reconstruction as one informative projection of a richer evolutionary state space. Lineage history, family-level organization and constraint similarity describe complementary aspects of evolutionary structure, and none of them alone is sufficient to characterize the full dynamics of gene family evolution.

Looking forward, this conceptual shift suggests that future evolutionary genomics will increasingly focus on family-level organization, cross-species comparability and representation-based descriptions of evolutionary constraints. Integrating homologous gene group ontology, bi-state evolutionary dynamics and high-dimensional representations provides a foundation for systematically investigating how complex gene families emerge, diversify and maintain functional coherence across deep evolutionary timescales.

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